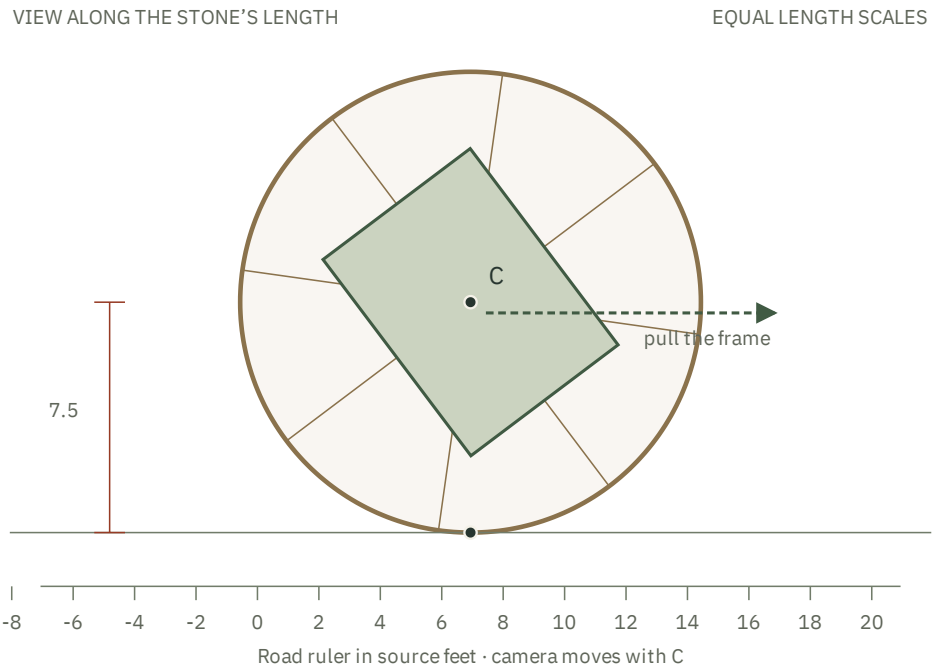
# Before the column stood.

The stone inside wheels. Rectangular comparison width 8, height 6; circular radius 7.5; rotation 53°. Lengths in source feet. The Paconius preset uses the reported 8 × 6 end section and the usual diameter reading of the 15-foot wheels.

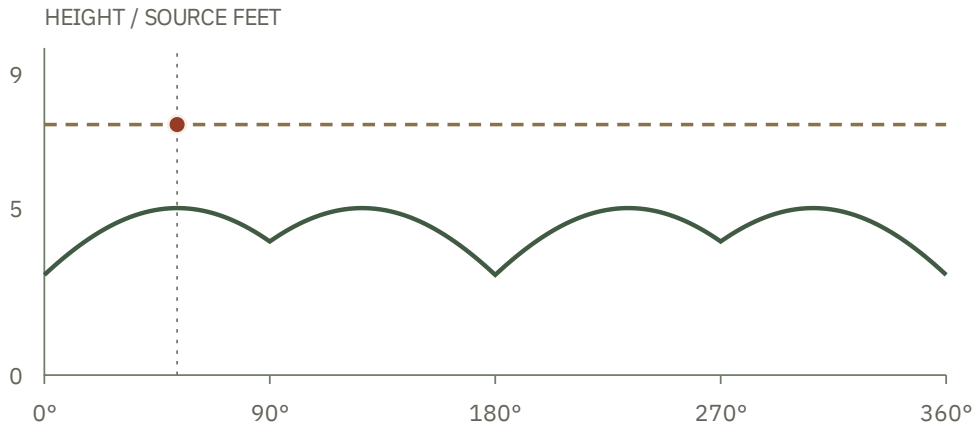
Edition b83ea0b6d6ac488ec6ce9d721aaf1a5a4dc996a8de6f57f17f3e1d6236d6eb01. Reopen these settings online. The reading, fonts and complete printed pages below open offline.

## Turn the stone




The centre stays at 7.5 source feet. A complete turn advances 47.123889804, the rolling circumference.

At this rotation: centre advance 6.937683777, centre height 7.5. Minimum full-turn clearance 2.5. The section clears the ground through the turn. Current lowest stone point 2.50001289.




Rotation · horizontal and vertical axes have different units

The green curve is the rectangular stone's required centre height. The dashed line is the chosen circular radius. A wheel radius below the curve cannot carry that section through a complete turn without ground intersection. Graph axes have different units; the end view preserves equal length scales.

# A fit is not a successful journey

With the Paconius readings, the half-diagonal is five source feet and the wheel radius seven and a half, leaving two and a half at the tightest rotation. This clearance test does not explain the reported veering. The source gives no complete steering geometry, friction measurements or rope-force history.

# Read the Latin

**X.2.11 · Free to turn**

## cnodaces autem in armillis inclusi liberam habuerunt versationem.

The pivots, enclosed in the rings, could turn freely.

The stone rotates while the timber frame is pulled. The next sentence explicitly supplies oxen; free rotation is not perpetual motion. Printed p. 230.

X.2.12 · The beam becomes the axle

## epistylia vero inclusa uti axes in rotis eadem ratione, qua scapi, sine mora ad opus pervenerunt.

The architraves, enclosed like axles in the wheels, reached the work without delay in the same way as the shafts.

The preceding page gives wheels of about twelve feet. The usual diameter reading follows Morgan's translation. No architrave cross-section is specified; the initial 8 × 6 section here is a chosen illustration. Printed p. 231.

---

X.2.12 · The route is part of the method

## nec ullus est clivus sed perpetuus campus.

There is no hill, but an uninterrupted plain.

Vitruvius also qualifies the distance as eight Roman miles. The drawing assumes a flat road and makes no conversion of his units or reconstruction of that route. Printed p. 231.

---

X.2.13 · The reported replacement block

## haec autem basis erat longa pedes duodecim, lata pedes VIII, alta pedes sex.

This base was twelve feet long, eight feet wide and six feet high.

The Paconius preset takes the twelve-foot length along the wheel axis, leaving an 8 × 6 end section. X.2.14 gives wheels of about fifteen feet, read as diameter here. Those are explicit modeling interpretations, not new archaeological measurements. Printed p. 231.

---

X.2.14 · Turning did not ensure a straight journey

## sed non poterat ad lineam via recta ducere, sed exibat in unam partem.

But he could not guide it straight along the road; it kept going off to one side.

Vitruvius reports that Paconius pulled a rope wound around the crossbars, repeatedly had to draw the machine back, and lost money. This two-dimensional model has no steering or rope-pull dynamics; it cannot establish the cause of that failure. Printed p. 231.

---

# Keep the grammar notes

## cnodaces — pivots

Nominative plural, the subject of habuerunt. Vitruvius uses a Greek technical loan; compare κνώδαξ, a pin or pivot. The diagram marks their common axis at C.

## armillis — in the rings

Ablative plural of armilla after in: location inside the bearing rings. The pivots turn within them; these rings are not the large road wheels.

## versationem — turning, rotation

Accusative singular, with liberam: the freedom to turn. Habuerunt is past-tense plural, agreeing with cnodaces.

## epistylia — architraves

Neuter nominative plural, the subject of pervenerunt. Inclusa agrees with it. Compare Greek ἐπιστύλιον: the horizontal member over columns.

## axes — axles

Plural of axis. Uti introduces the comparison: the architraves themselves occupy the axle position inside the wheels.

## rotis — in the wheels

Ablative plural of rota after in. These are the large rolling wheels, distinct from the bearing rings named armillae.

## clivus — a slope or hill

Nominative singular. Nec ullus denies any hill in this description of the route; the present model assumes that stated flatness.

## lata — wide

Feminine nominative singular, agreeing with basis. Longa, lata and alta give three different dimensions; width eight and height six make the end section used here.

## ducere — to lead or draw

An infinitive after poterat, could. The reported problem is guiding the machine along a straight line, not an assertion that its rectangular load would strike the road.

# Greek words carried into Latin

ὁ κνώδαξ, genitive κνώδακος — a pivot or pin on which something turns. Latin *cnodax*. The singular changes its stem before the genitive ending. The Latin passage has the plural cnodaces. LSJ via Logeion.

ὁ κύλινδρος, genitive κυλίνδρου — a roller or cylinder. Latin *cylindrus*. Vitruvius compares the method to the rollers used to level palaestra walks. His Latin plural is cylindri. LSJ via Logeion.

τό ἐπιστύλιον, genitive ἐπιστυλίου — an architrave, the member over columns. Latin *epistylium*. A neuter noun. Latin epistylia is plural in the selected sentence; these beams turn within the wheels. LSJ via Logeion.

These are Greek dictionary headwords explaining the Latin technical vocabulary, not a Greek original or back-translation of Vitruvius's account. Definitions and reading notes are new; the links identify LSJ entries in Logeion.

# Sources and limits

A modern kinematic reading of Vitruvius X.2.11–14, not an archaeological reconstruction or engineering plan. The road is flat, bodies rigid and wheels circular; rolling has no slip. Two end wheels are represented by one end view. The stone and wheels turn together. Actual bearings, frame dimensions, taper, soil, friction, rope winding, steering, forces, masses and animal effort are outside the model. The bare rectangle is an explicit modern counterexample, not a method prescribed in the passage.

Krohn's 1912 edition, printed pp. 230–231 (PDF pages 248–249), Internet Archive original. The complete apparatus and the gap in the timber-frame description remain visible. Line-end hyphens are joined in the short excerpts; Roman numerals are normalized in capitalization. No missing frame description is silently restored. Historical text and scans: public domain. Code, new diagrams and English: MIT, Anton Soloviev, 2026. Two JPEGs, two fonts and their two licenses were verified against SHA-256 identities before export.

quantum velit et dextra ac sinistra ab latere proclinando onus deponere potest.

Harum machinationum omnium, quae supra sunt scriptae, rationes non modo ad has res, sed etiam ad onerandas et exonerandas naves sunt paratae, aliae erectae, aliae planae in carchesiis versatilibus conlocatae. non minus sine tignorum erectionibus in plano etiam eadem ratione et temperatis funibus et trocleis subductiones navium efficiuntur.

11 Non est autem alienum etiam Chersiphronos ingeniosam rationem exponere. is enim scapos columnarum e lapidicinis cum deportare vellet Ephesi ad Dianae fanum, propter magnitudinem onerum et viarum campestrem mollitudinem non confisus carris, ne rotae devorarentur, sic est conatus. de materia trientali scapos quattuor, * * * * duos transversarios interpositos, quanta longitudo scapi fuerat, complectit et compegit et ferreos cnodacas uti subscudes in capitibus scaporum inplumbavit et armillas in materia ad cnodacas circumdandos infixit; cnodaces autem in armillis inclusi liberam habuerunt versationem. iamiam ita, cum boves ducerent subiuncti, scapi versando
12 in cnodacibus et armillis sine fine volvebantur. cum autem scapos omnes ita vexerant et instabant epistyliorum vecturae, filius Chersiphronos Metagenes transtulit ex scaporum vectura etiam in epistyliorum deductione. fecit enim rotas circiter pedum duodenûm et epistyliorum capita in medias rotas inclusit; item bucculis ligneis capita religavit; eadem

---

1 adlatere *x* (adlatera $G^c$) 6 carchesis *HS*, carthesis *G* 10 crestiphonos *x* 14 carris $SG^c$: caris *HG* 16 interpositos *ed*: interpostios *x* 17 fuerat *ed*: fuerit *x* complectit *rec*: -tet *G*, -t& *HS* cōnpeget *G* chodacas *HS*, codacas *G* 19 chodacas *x* chodaces *x* 21 iamiam *Kr*: tantam *x* subiuncti *G*: -tis *HS* scapi *Joc*: scapo *x* 22 codacibus *HG*, -tibus *S* 23 vexerant *Ro*: -runt *x* 24 cresiphonos *x* ex scaporum $S^c$: excaporum *x* 27 item — religavit *transp. Kr*: *post* infixit (*v. 19*) *exh. x sic*: item bucculis (*S*, bacculis $HS^c$, buculis *G*, baculis *in marg.* $G^c$) ligneis ($G^c$, tigneis *HS*, digneis *G*) capita religavit

Vitruvius X.2, printed p. 230; PDF page 248. Source scan, including the apparatus.

ratione cnodaces in capitibus armillis inclusit: ita cum trientes a bubus ducerentur, in armillis inclusi cnodaces versabant rotas, epistylia vero inclusa uti axes in rotis eadem ratione, qua scapi, sine mora ad opus pervenerunt. exemplar autem erit eius, quemadmodum in palaestris cylindri exaequant ambulationes. neque hoc potuisset fieri, nisi primum propinquitas esset — non enim plus sunt ab lapidicinis ad fanum milia passuum octo — nec ullus est clivus sed perpetuus campus.

Nostra vero memoria cum colossici Apollinis in fano 13
basis esset a vetustate diffracta, [et] metuentes, ne caderet ea statua et frangeretur, locaverunt ex eisdem lapidicinis basim excidendam. conduxit quidam Paconius. haec autem basis erat longa pedes duodecim, lata pedes VIII, alta pedes sex. quam Paconius gloria fretus non uti Metagenes adportavit, sed eadem ratione alio genere constituit machinam facere. rotas enim circiter pedum XV fecit et in 14
his rotis capita lapidis inclusit, deinde circa lapidem fusos sextantales ab rota ad rotam ad circinum compegit, ita uti fusus a fuso non distaret pedem [esse] unum. deinde circa fusos funem involvit et bubus iunctis funem ducebant. ita cum explicaretur, volvebat rotas, sed non poterat ad lineam via recta ducere, sed exibat in unam partem. ita necesse erat rursus retroducere. sic Paconius ducendo et reducendo pecuniam contricavit, ut ad solvendum non esset.

Pusillum extra progrediar et de his lapidicinis, quem- 15
admodum sint inventae, exponam. Pixodarus fuerat pastor. is in his locis versabatur. cum autem cives Ephesiorum cogitarent fanum Dianae ex marmore facere decernerent-

---

1 chodaces *x* in capitibus armillis *Kr*: et armillas in capitibus *x* 2 triantes *S* chodaces *x* 3 versabant *ed*: -bant̃ *HG*, -bantur *S* 4 rationem *a. c. H*, -nē *a. ras. S* 7 ess& *G*: esse *H*, e͞e *S* 8 lapidicinis $G^c$: -nes *x* 9 est *om. G* 11 *del. Ro* caderet *S*: cederet *HG* 14 XII *GS* 19 sextantales *Joc*: sextantes *x* 20 *del. rec* (*cf. 218, 4 etc.*) 21 et *ed*: ut *x* 24 paconius $G^c$: paconius *HG*, peonius *S* 25 *fort.* solvendo *cum* ad *additivo*

Vitruvius X.2, printed p. 231; PDF page 249. Source scan, including the apparatus.